Delta Hedging in Discrete Time:

Rebalancing Error, Transaction Costs and Volatility Misspecification

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Abstract

In the Black–Scholes model a European call can be replicated exactly by trading continuously in the underlying stock. This note studies, by Monte Carlo simulation, what remains of this result under three realistic departures: the hedge is rebalanced at finitely many dates, every trade incurs a proportional cost, and the volatility used to hedge differs from the one realised by the market. For each case I derive the leading-order behaviour of the hedger's profit and loss with an elementary Itô–Taylor argument and compare it with simulations. The rebalancing error decays like $N^{-1/2}$ in the number N of rebalancing dates; with costs, the expected loss grows like $N^{1/2}$, which creates an optimal rebalancing frequency, and Leland's adjusted volatility removes almost all of the cost bias; with a misspecified volatility, the profit is driven by the gamma-weighted gap between implied and realised variance, which I verify path by path. All code is available at `https://github.com/USERNAME/delta-hedging`.

1 Introduction

The Black–Scholes price of an option is the initial cost of a self-financing portfolio that replicates its payoff [2]. Replication requires trading continuously, in a frictionless market, with knowledge of the true volatility. None of these holds in practice, so a trader who sells an option at its Black–Scholes price and hedges it is exposed to a residual profit and loss (P&L).

This note quantifies that residual in a controlled setting. The stock follows a geometric Brownian motion; a trader sells a three-month at-the-money call and delta-hedges it. Three questions are studied, one per section:

1. How large is the hedging error when the hedge is rebalanced only N times? (Section 3)
2. How do proportional transaction costs change the picture, and can they be priced in? (Section 4)
3. What happens when the trader hedges with a volatility different from the realised one? (Section 5)

The answers are classical [6, 4, 5, 1, 3]; the purpose here is to re-derive the mechanisms in a unified way and to check each prediction numerically.

2 Setting

Model. Under the real-world probability the stock satisfies

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t, \tag{1}$$

and cash earns the constant rate r. The Black–Scholes price of a call with strike K and maturity T is

$$C(t,S) = S\,\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad d_{1,2} = \frac{\ln(S/K) + (r \pm \tfrac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad \tau = T - t,$$

with delta $\Delta = \partial_S C = \Phi(d_1)$ and gamma $\Gamma = \partial_{SS} C = \varphi(d_1)/(S\sigma\sqrt{\tau})$. The price solves the Black–Scholes equation

$$\partial_t C + \tfrac{1}{2}\sigma^2 S^2\,\Gamma + rS\,\Delta - rC = 0. \tag{2}$$

Throughout, $S_0 = K = 100$, $T = 0.25$, $r = 2\%$, $\mu = 8\%$ and $\sigma = 20\%$, so that $C(0,S_0) = 4.23$.

Hedging strategy. The trader sells the call at time 0 for $C(0,S_0)$, buys Δ_0 shares and keeps the rest in cash. At the dates $t_i = iT/N$, $i = 1,\dots,N-1$, the position is adjusted to Δ_{t_i} shares, the trade being financed from cash; no money is added or withdrawn. At maturity the trader pays $(S_T - K)^+$. The terminal P&L is

$$\text{P\&L} = B_T + \Delta_{t_{N-1}} S_T - (S_T - K)^+,$$

where B_T is the cash account. Under continuous rebalancing in the model (1), P&L $= 0$ almost surely.

Simulation. Paths are simulated exactly on the grid, $S_{t_{i+1}} = S_{t_i} \exp\left((\mu - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\,Z_i\right)$ with Z_i i.i.d. standard normal and $\Delta t = T/N$, so the only approximation is the discreteness of the hedge. Results use 20,000 paths for the main figures and 10,000 paths per point in the parameter sweeps.

3 Discrete rebalancing

Local error. Consider one period $[t_i, t_{i+1}]$ and write $\delta S = S_{t_{i+1}} - S_{t_i}$. To second order, the option moves by $\partial_t C\,\Delta t + \Delta\,\delta S + \tfrac{1}{2}\Gamma\,\delta S^2$, the shares by $\Delta\,\delta S$, and the cash, which equals $C - \Delta S$ when the portfolio is balanced, earns $r(C - \Delta S)\Delta t$. Using (2) to eliminate $\partial_t C$, the change in the hedged position of the option seller is

$$\varepsilon_i \approx \tfrac{1}{2}\,\Gamma_{t_i} S_{t_i}^2 \left(\sigma^2 \Delta t - \left(\frac{\delta S}{S_{t_i}}\right)^2\right) \approx \tfrac{1}{2}\,\Gamma_{t_i} S_{t_i}^2\,\sigma^2 \Delta t\,\left(1 - Z_i^2\right). \tag{3}$$

The delta terms cancel exactly: this is why the drift μ plays no role. What is left is a convexity term, of zero mean, because $\mathbb{E}[Z_i^2] = 1$, and of variance $\tfrac{1}{2}(\Gamma S^2 \sigma^2 \Delta t)^2$, because $\mathrm{Var}(Z^2) = 2$.

Global error. The local errors are uncorrelated, so their variances add:

$$\mathrm{Var}(\text{P\&L}) \approx \tfrac{1}{2}\sum_i \mathbb{E}\left[(\Gamma_{t_i} S_{t_i}^2 \sigma^2)^2\right] \Delta t^2 \approx \frac{T}{2N} \int_0^T \mathbb{E}\left[(\Gamma_t S_t^2 \sigma^2)^2\right]\,\mathrm{d}t.$$

The standard deviation of the P&L therefore decays like $N^{-1/2}$. For an at-the-money option, Derman and Kamal [4] obtain the explicit approximation

$$\mathrm{Std}(\text{P\&L}) \approx \sqrt{\frac{\pi}{4}}\,\frac{\sigma\,\mathcal{V}}{\sqrt{N}}, \qquad \mathcal{V} = \partial_\sigma C(0,S_0). \tag{4}$$

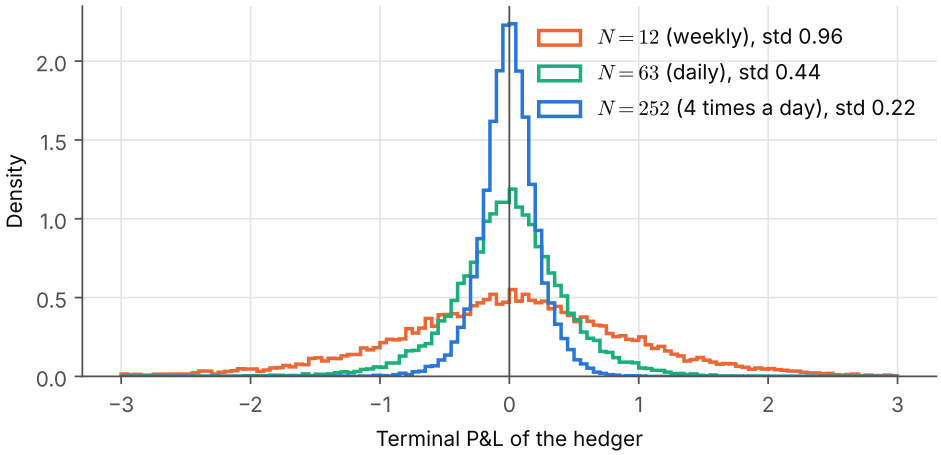


Figure 1: Distribution of the terminal P&L of a delta-hedged short call for three rebalancing frequencies. The option is worth 4.23.

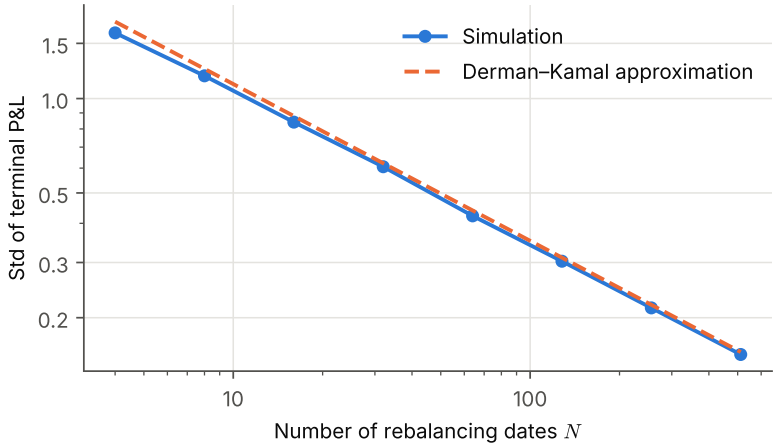


Figure 2: Standard deviation of the P&L against N (log–log scale), with the Derman–Kamal approximation (4).

Results. Figure 1 shows the P&L distribution for weekly ($N = 12$), daily ($N = 63$) and four-times-daily ($N = 252$) rebalancing. The distributions are centred at zero (means -0.009, -0.001 and $+0.001$) with standard deviations 0.96, 0.44 and 0.22: multiplying N by four roughly halves the error. Figure 2 confirms the rate: the fitted log–log slope is -0.489, against -0.5 in theory, and (4) is within 8% of the simulation over the whole range.

Role of the drift. The cancellation in (3) only holds to leading order. Hedging under drifts of -20% and $+40\%$, the mean P&L moves by at most 0.028 with daily rebalancing and by at most 0.0007 with $N = 1000$: the drift affects the hedge only through discretisation effects, which vanish as $N \to \infty$.

4 Transaction costs

Cost model. Each trade of n shares at price S now costs $\frac{k}{2}|n|S$, where k is the round-trip proportional cost; I take $k = 0.4\%$. All rebalancing trades, including the initial purchase of Δ_0 shares, are charged.

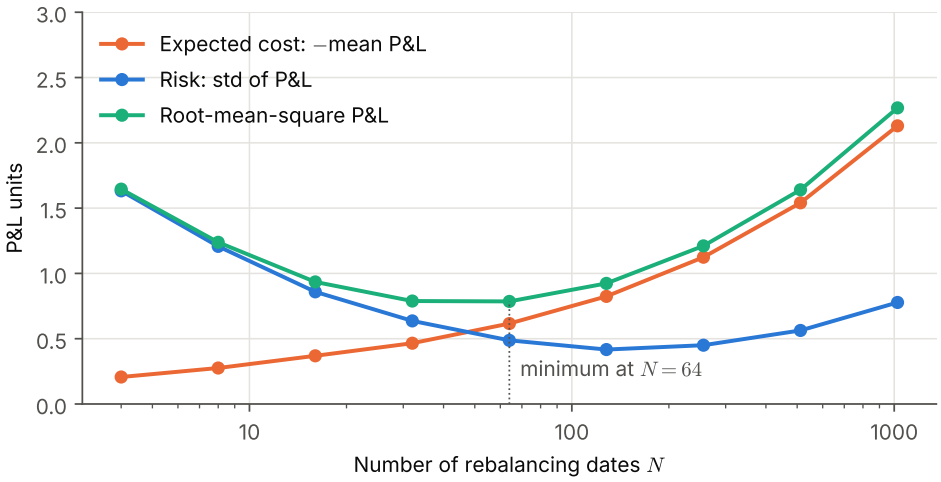


Figure 3: Hedging with transaction costs ($k = 0.4\%$ round trip): expected cost, risk, and root-mean-square P&L against the number of rebalancing dates.

Scaling of the cost. Over one period the hedge ratio changes by $\delta\Delta \approx \Gamma\,\delta S \approx \Gamma S\sigma\sqrt{\Delta t}\,Z$. Since $\mathbb{E}|Z| = \sqrt{2/\pi}$, the expected cost of one rebalance is

$$\mathbb{E}[\text{cost}_i] \approx \frac{k}{2}\sqrt{\frac{2}{\pi}}\,\sigma\,\Gamma_{t_i} S_{t_i}^2 \sqrt{\Delta t}, \tag{5}$$

and the total expected cost, a sum of N such terms, grows like $N\sqrt{\Delta t} \propto \sqrt{N}$. Frequent rebalancing reduces the hedging error of Section 3 but increases the bill: costs grow like $\sqrt{N}$ while the error shrinks like $1/\sqrt{N}$.

Results. Figure 3 shows the trade-off. The expected loss increases steadily with N and reaches 2.13, half the option's value, at $N = 1024$. The standard deviation is no longer monotone: it falls to 0.42 at $N = 128$ and then rises again, because the random amount of costs paid becomes itself a source of risk. The root-mean-square P&L, which penalises both bias and risk, is minimised at $N = 64$, roughly daily rebalancing. Exact replication is no longer possible, and selling at the Black–Scholes price is a losing strategy on average.

Leland's correction. Equation (5) has the same form as the convexity term in (3): both are proportional to ΓS^2. The cost can therefore be absorbed into an enlarged variance. Requiring the extra convexity premium $\frac{1}{2}\Gamma S^2(\hat{\sigma}^2 - \sigma^2)\Delta t$ to pay for the expected cost (5) gives Leland's volatility [6]

$$\hat{\sigma}^2 = \sigma^2\left(1 + \sqrt{\frac{2}{\pi}}\,\frac{k}{\sigma\sqrt{\Delta t}}\right). \tag{6}$$

The trader sells the option at the Black–Scholes price computed with $\hat{\sigma}$ and hedges with the corresponding delta. For daily rebalancing, $\hat{\sigma} = 22.4\%$ and the option is sold for 4.71 instead of 4.23.

Figure 4 compares the two strategies. With Leland's volatility the mean P&L stays between -0.111 and -0.093 for every N. This residual is the cost of buying the initial hedge, $\frac{k}{2}\Delta_0 S_0 = 0.108$, a one-off trade that (6), built from rebalancing costs, does not price. The risk also keeps decreasing with N, to 0.13 at $N = 1024$ against 0.78 for the Black–Scholes hedge. The approximation behind (6) requires $k/(\sigma\sqrt{\Delta t})$ to remain moderate; as $\Delta t \to 0$ with k fixed, $\hat{\sigma}$ explodes, and it is known that perfect hedging under proportional costs is impossible in that limit.

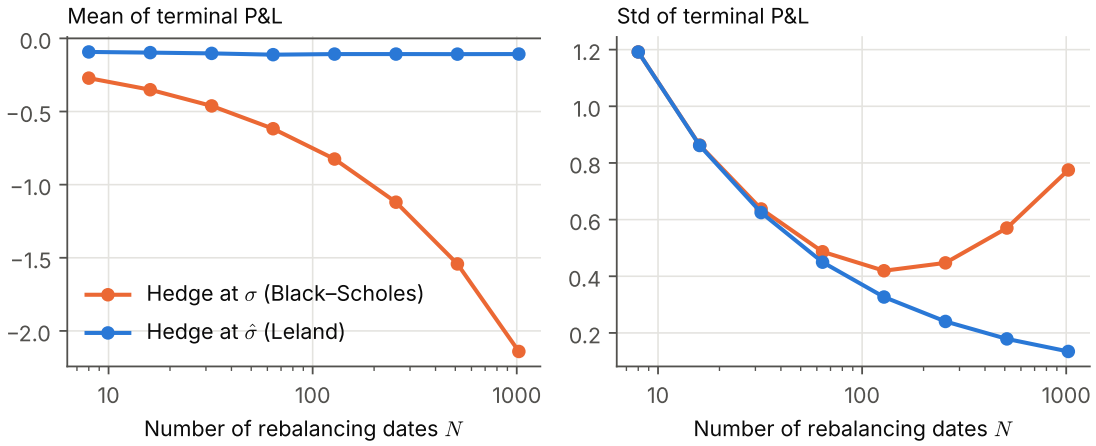


Figure 4: Mean (left) and standard deviation (right) of the P&L under transaction costs, when pricing and hedging with the true volatility σ or with Leland's volatility $\hat{\sigma}$.

5 Hedging with the wrong volatility

In practice the volatility is not observed. Options trade at an *implied* volatility Σ, while the stock realises some volatility σ that is only known after the fact. Here the stock realises $\sigma = 15\%$ while the option is sold at $\Sigma = 20\%$. Costs are set to zero.

Hedging at the implied volatility. Let C^Σ, Δ^Σ, Γ^Σ denote the Black–Scholes quantities computed with Σ. Repeating the computation of Section 3 in continuous time, but using the Black–Scholes equation (2) at volatility Σ while S has quadratic variation $\sigma^2 S^2\,\mathrm{d}t$, Itô's formula gives for the seller's hedged position Π

$$\mathrm{d}\Pi_t - r\Pi_t\,\mathrm{d}t = \tfrac{1}{2}\,\Gamma_t^\Sigma S_t^2\left(\Sigma^2 - \sigma^2\right)\mathrm{d}t,$$

so that

$$\mathrm{P\&L} = \frac{\Sigma^2 - \sigma^2}{2}\int_0^T e^{r(T-t)}\,\Gamma_t^\Sigma S_t^2\,\mathrm{d}t. \tag{7}$$

This is the classical robustness formula of El Karoui, Jeanblanc-Picqué and Shreve [5]. Since $\Gamma > 0$ for a call, the seller makes money on *every* path when the implied volatility exceeds the realised one, but the amount depends on the path: it is large when the stock stays close to the strike near maturity, where gamma is concentrated.

Hedging at the realised volatility. Alternatively, the trader sells at C^Σ but hedges with Δ^σ. The hedge then replicates exactly a call priced at σ, and the trader keeps the price difference:

$$\mathrm{P\&L} = e^{rT}\left(C^\Sigma(0,S_0) - C^\sigma(0,S_0)\right) = 0.997. \tag{8}$$

The profit is deterministic, but the strategy requires knowing σ in advance.

Results. The left panel of Figure 5 plots, path by path, the simulated P&L of the implied-volatility hedge against the prediction (7), evaluated on the same path. With $N = 1000$ the correlation is 0.978, and the residual, which is the discrete-rebalancing error of Section 3, has standard deviation 0.070. The P&L is positive on all 20,000 paths, ranging from 0.16 to 1.92.

The right panel compares the two strategies. Both earn about the price gap on average (0.988 and 0.997, against 0.997 from (8)), but their risks differ: 0.334 for the implied hedge

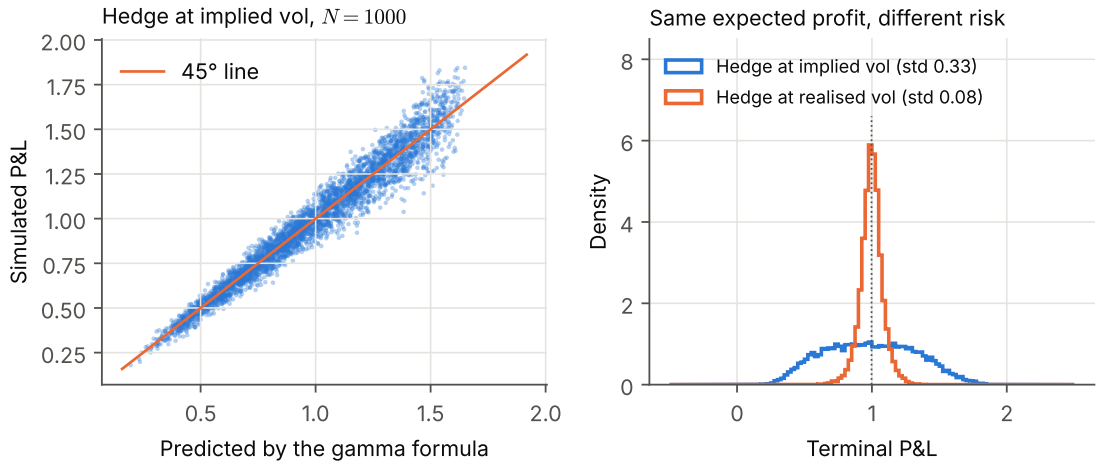


Figure 5: Option sold at implied volatility $\Sigma = 20\%$, stock realising $\sigma = 15\%$. Left: simulated P&L of the implied-volatility hedge against formula (7), on 4,000 paths. Right: P&L of the implied-volatility and realised-volatility hedges; the dotted line is the locked-in profit (8).

	Hedge at implied Σ		Hedge at realised σ	
N	mean	std	mean	std
16	0.986	0.703	0.992	0.627
64	0.987	0.447	0.994	0.318
256	0.987	0.362	0.999	0.164
1024	0.995	0.335	0.997	0.083

Table 1: P&L of the two hedging strategies when $\Sigma = 20\%$ and $\sigma = 15\%$ (10,000 paths per row). Only the realised-volatility hedge converges to a deterministic profit.

against 0.083 for the realised one. Table 1 shows that the second number tends to zero as N grows, while the first levels off. Its randomness is not a discretisation artefact; it is intrinsic to (7).

6 Conclusion

Each departure from the Black–Scholes assumptions leaves a residual whose leading-order behaviour follows from the same second-order expansion. Discrete rebalancing creates an error of order $N^{-1/2}$; proportional costs create a bias of order $N^{1/2}$, hence an optimal rebalancing frequency, and Leland's volatility prices this bias in almost entirely; a misspecified volatility creates a profit driven by the gamma-weighted gap between implied and realised variance. In every case the simulations agree with the leading-order formulas.

Two directions seem natural to pursue. First, Leland's correction is an approximation: with transaction costs, the exact superhedging price solves a *nonlinear* pricing equation, which can be written as a backward stochastic differential equation (BSDE) [3]. Solving such equations numerically, and comparing the result with Leland's formula, would make the analysis of Section 4 exact rather than asymptotic. Second, the hedging dates were fixed in advance; rebalancing only when the delta leaves a band around its target is a simple way to trade fewer shares for the same risk, and comparing it with calendar hedging would refine the trade-off of Figure 3.

Reproducibility. Every figure and every number in this note is produced by a single script, `experiments/run_experiments.py`, which writes the numbers into the LaTeX source. The code, with unit tests, is available at `https://github.com/USERNAME/delta-hedging`.

References

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